

Rapid Quantitative Analysis of Lipids Using a Colorimetric Method in a Microplate Format

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Abstract A colorimetric sulfo-phospho-vanillin (SPV) method was developed for high throughput analysis of total lipids. The developed method uses a reaction mixture that is maintained in a 96-well microplate throughout the entire assay. The new assay provides the following advantages over other methods of lipid measurement: (1) background absorbance can be easily corrected for each well, (2) there is less risk of handling and transferring sulfuric acid contained in reaction mixtures, (3) color develops more consistently providing more accurate measurement of absorbance, and (4) the assay can be used for quantitative measurement of lipids extracted from a wide variety of sources. Unlike other spectrophotometric approaches that use fluorescent dyes, the optimal spectra and reaction conditions for the developed assay do not vary with the sample source. The developed method was used to measure lipids in extracts from four strains of microalgae. No significant difference was found in lipid determination when lipid content was measured using the new method and compared to results obtained using a macro-gravimetric method.

Keywords Sulfo-phospho-vanillin · Microalgae · Macro-gravimetric

Abbreviations

SPV Sulfo-phospho-vanillin
ATCC American type culture collection
UTEX The culture collection of algae

ANCOVA Analysis of covariance
HSD Honestly significant differences
vvm Volume per volume per minute

Introduction

Lipids are an important group of compounds that provide several biological functions such as energy storage, cell membrane structure and signaling [1, 2]. For this reason, lipid analyses are performed routinely in many different research areas. For example, the screening of oleaginous organisms has extensive application in both research and industry settings for identifying and producing food supplements and renewal biofuels [3–5]. In addition, in aquaculture processes analysis of lipid content in animals and microorganisms assists with monitoring the health conditions of the culture and developing growth strategies [6, 7]. Several methods have been developed to quantify total lipids. The most common approach is a macro-gravimetric method in which lipids are extracted from a sample, the extraction solvent is evaporated and the retained material is measured as the lipid content [8, 9]. This traditional gravimetric method requires a relatively large quantity of sample and is time-consuming and labor-intensive when analysis of many samples is needed. Spectrofluorometric analysis of lipid, which uses the fluorescent dye Nile red, was originally developed by Greenspan et al. [10] and has also been modified for quantification of total lipids [11–14]. While this approach is high-throughput, environmental factors and other components in the cell cytoplasm, such as proteins and pigments, interfere with the assay and the fluorescence intensity varies between samples [14–16]. For

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this reason accurate quantification of lipids using this approach requires that the optimal spectra and reaction conditions be determined for each type of sample prior to fluorescent measurements [13, 17].

The colorimetric sulfo-phospho-vanillin (SPV) method developed by Chabrol et al. [15, 18] is an attractive alternative for lipid measurement because of its fast response and relative ease in sample handling. The SPV method has been modified for diverse applications such as the determination of total lipids in serum, food and ecological samples [16, 19–23]. A micro scale modification of the SPV assay was developed by Van Handel [24] for determination of total lipids in a single mosquito, and assessed by several investigators as a more time and labor efficient approach compared to gravimetric methods [25–27]. In Inouye's report [25], 0.25 ml sample, 0.1 ml sulfuric acid and 2.4 ml vanillin reagent (1.2 mg vanillin per ml 68% phosphoric acid) were required for the micro-colorimetric assay. Similarly, in a report published by Lu et al. [27] 0.1 ml sample, 2.5 ml sulfuric acid and 5 ml vanillin reagent were needed to conduct the SPV assay on a micro scale. Both methods were completed in 13 × 100 mm culture tubes. Transfer of reactants to proper containers was required for absorbance measurement. Since color develops continuously, careful sample handling and control of color development are critical using this micro-scale approach.

The present work reports on an adaptation of the SPV method for completion of lipid quantification in a 96-well microplate for higher throughput and reduced costs. The adapted method involves an assay in which the reagent mixture is confined to one microplate for the entire assay. This enables faster measurement of multiple samples with easy background correction and more consistent monitoring of color development. As an example, total lipids in extracts from microalgae, which contain a dark green background, was successfully measured using this assay approach with corn oil as a standard.

Materials and Methods

Materials

Concentrated sulfuric acid, *o*-phosphoric acid (85%), chloroform and methanol, all ACS grade, were purchased from Thermo Fisher Scientific Inc (Waltham, MA, USA). Vanillin (≥98%) was purchased from Sigma-Aldrich (St. Louis, MO, USA). Commercial corn oil, canola oil, flax oil, sunflower oil and cod liver oil were obtained from local markets. A cholesterol standard for clinical work was purchased from MP biomedical (Solon, OH, USA). Flat

bottom, polystyrene 96-well microplates (Costar 3370) were obtained from Corning Incorporated (Corning, NY, USA).

Preliminary Assessment of General Assay Conditions

A preliminary microplate test was performed to determine the initial assay conditions. The general assay steps followed are presented in Fig. 1. The volume of reagents per well in the preliminary study included 100 µl of concentrated sulfuric acid and 100 µl vanillin–phosphoric acid reagent. Three concentrations of vanillin were examined (1, 0.5, and 0.1 mg vanillin per ml 68% phosphoric acid) to determine the suitable concentration for assay in a microplate format. Standard samples were prepared by mixing corn oil in solvent (chloroform:methanol = 1:1) at 15 mg/ml. Standards were added into each well of the microplate by varying the volume (0.5–30 µl) and then 100 µl concentrated sulfuric acid was added after the solvent was evaporated at 90 °C for about 10 min. The microplate was then incubated on a dry heating bath (Isotemp 125D, Thermo Fisher Scientific Inc.) at 90 °C for 10 min and cooled to room temperature on ice water (~2 min). A microplate reader (Model VMax, Molecular Device, Sunnyvale, CA, USA) with compatible software, Softmax v 2.43, was used to measure background absorbance. Then, 100 µl vanillin–phosphoric acid reagent was added for

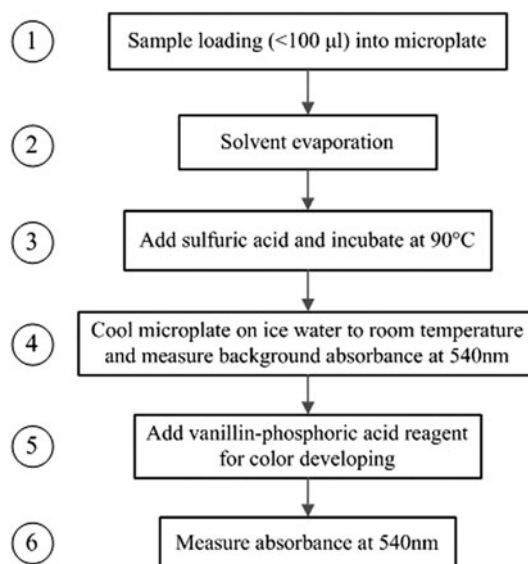


Fig. 1 General analysis procedures of SPV method in a microplate. Volumes of sulfuric acid and incubation times in step 3, and vanillin–phosphoric acid reagent levels and color development time in step 5 were varied to determine the ideal assay conditions for high throughput microplate analysis of lipid

color development. Absorbance at 540 nm was measured after 5 min of color development.

Examination of Solvent Ratios for Sample Preparation

Solvent mixtures including four different ratios of chloroform and methanol (chloroform only, chloroform:methanol = 2:1, 1:1 and 1:2) were tested to determine the effect of sample preparation on the assay. Samples containing corn oil in solvent mixtures (15 mg corn oil per ml solvent) were pipetted into the microplate wells by varying the volume from 1 to 15 μ l. The solvent was evaporated at 90 °C ($\sim$ 10 min), and then 100 μ l concentrated sulfuric acid was added. The microplate was then incubated on a dry heating bath at 90 °C for 20 min. Absorbance at 540 nm was measured as background after the microplate was cooled to room temperature on ice water. Then, 100 μ l vanillin–phosphoric acid reagent (0.5 mg vanillin per ml 68% phosphoric acid) was added for color development. Absorbance at 540 nm was measured after 5 min of development.

Examination of Vanillin and Phosphoric Acid Concentrations

After the sample preparation method was determined, concentrations of vanillin and phosphoric acid were investigated to determine their influence on the assay and if lower concentrations could be used in a high-throughput format. All assays used 100 μ l vanillin–phosphoric acid reagent for color development. Four vanillin concentrations in 68% phosphoric acid were first tested at 0.5, 0.375, 0.25 and 0.125 mg/ml. After an ideal concentration of vanillin (0.25 mg/ml) was determined, four different concentrations of phosphoric acid (17, 34, 51 and 68%) were tested. To examine the possibility of further reducing vanillin in the assay, the vanillin concentration was optimized after the suitable concentration of phosphoric acid had been determined.

Volume of Vanillin Phosphoric Acid Reagent and Concentrated Sulfuric Acid

In an attempt to reduce water use and final volume, different volumes of vanillin phosphoric acid reagent were tested. Four volumes of vanillin phosphoric acid reagent (20, 25, 50, 100 μ l) containing an equivalent amount of vanillin were first investigated to determine whether the volume of the reagent could be reduced by increasing the vanillin concentration. After the vanillin concentration in the phosphoric acid reagent and volume of vanillin phosphoric reagent in the assay were determined, the effect of

concentrated sulfuric acid volume (100, 150 and 200 μ l) on digestion was tested.

Examination of Reaction Time and Stability of Absorbance Linearity

To investigate the effect of reaction time on the assay, standard samples (10 mg corn oil per ml solvent) were loaded at varying volumes (12, 8, 4, 1 μ l) in the assay with 100 μ l concentrated sulfuric acid. The reaction time was varied from 10 to 60 min by carefully transferring samples from the original microplate to a new one at 10 min intervals. The new microplate was maintained on ice water until all the samples were transferred. After all samples had been cooled and background absorbance had been measured, 50 μ l of vanillin–phosphoric acid reagent (0.2 mg vanillin per ml 17% phosphoric acid) was added to each well. To test the stability of absorbance linearity, absorbance measurements made at 540 nm were recorded from 5 min to 3.5 h after vanillin–phosphoric acid reagent addition.

Relative Absorbance and Linearity of Different Lipid Samples

To survey the applicability of the assay, six different lipid samples, including corn oil, canola oil, flax oil, sunflower oil, cod liver oil and cholesterol, were tested in the range 5–150 μ g by varying loading volume of standard solutions (10 mg lipid per ml solvent). After the solvent was evaporated, 100 μ l concentrated sulfuric acid was added to each well and then the microplate was incubated at 90 °C for 20 min. 50 μ l of vanillin–phosphoric acid reagent (0.2 mg vanillin per ml 17% phosphoric acid) was added to each well for color development. Absorbance at 540 nm was measured after 10 min of development.

Applicability of the Assay for Measuring Lipid Extracted from Microalgae

Four different strains of microalgae, *Chlorella vulgaris* UTEX 259, *Chlorella sorokiniana* UTEX 2805, *Chlorella minutissima* UTEX 2341 and *Chlorella* sp. NC64A were obtained from The Culture Collection of Algae (UTEX) and American Type Culture Collection (ATCC). *Chlorella* UTEX 259 and UTEX 2805 were cultured in inorganic N8 medium, and *Chlorella* UTEX 2341 and NC64A were cultured in N8Y medium prepared by adding 0.1% yeast extract in N8 medium. All strains were cultured at 25 °C with either ambient air supply at 1 vvm or 2% CO₂ at 0.5 vvm in 1-l glass bottles and irradiated with fluorescent light at $\sim$ 2,000 Lux on a 16:8 h light/dark cycle. Cultures were harvested in the late log phase of growth, centrifuged

at 2,000g and the retained pellet was lyophilized. The lipid extraction and purification methods were according to Folch [9] with slight modifications. Samples of purified lipid extracts were mixed with twice the volume of methanol for the colorimetric assay in a microplate, and the results were compared with the total lipid measured using the macro-gravimetric method [25].

Data Analysis

Linearity was determined by plotting absorbance versus lipid amount in the assay and examining the R^2 value upon linear regression of the data. Analysis of covariance (ANCOVA) and Tukey's HSD test were used to assess differences in the mean slope obtained upon linear regression of absorbance versus lipid amount. A paired t test and a graphical statistic paired test based on Bland's method [28] were performed to evaluate the agreement between the developed method and the macro-gravimetric method when lipid samples from microalgae were analyzed. Paired t tests, ANCOVA and Tukey's HSD tests were performed using JMP IN v.8.0 (SAS Institute Inc. Cary, NC, USA).

Results and Discussion

Preliminary Test of Assay Conditions

A preliminary microplate test was performed to determine a suitable range of vanillin in the assay. All tested concentrations (1, 0.5, and 0.1 mg vanillin per ml 68% phosphoric acid) gave good linearity of absorbance ($R^2 > 0.95$) at 540 nm in the range of 15–120 μg lipid (Data not shown). Vanillin concentration at 0.5 mg/ml gave a sufficient and relatively linear result, and therefore was selected as the starting point for optimization. Although the preliminary test showed very good linearity, there were some samples with noticeable film development and corrosion in the bottom of the wells of the microplate that might interfere with the reaction between lipid and sulfuric acid and affect color development. Therefore, different solvent ratios were also included in the development of the method.

Development of Reaction Conditions

Sample Preparation in Different Solvent Combinations

Only the samples prepared in a chloroform:methanol ratio of 2:1 had clear reactions and differences in color development (Fig. 2). In some of the tests using chloroform:methanol ratios of 2:1 and 1:1, corrosion and film

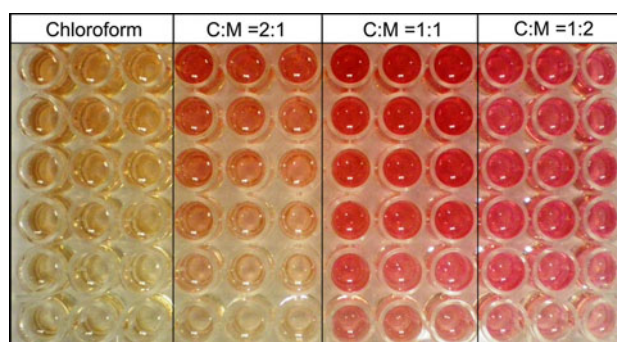


Fig. 2 Influence of solvent combinations on reaction and color development. *C* chloroform, *M* methanol. Samples (corn oil in solvent, 15 mg/ml) were added from 1 to 15 μl and incubated with 100 μl concentrated H_2SO_4 at 90 $^\circ\text{C}$ for 20 min. 100 μl vanillin–phosphoric acid reagent (0.5 mg vanillin per ml 68% phosphoric acid) was added for color development. Absorbance measurements were made after 5 min of color development. Sample loading volumes from top to bottom are 15, 12, 9, 6, 3, and 1 μl

development occurred in the bottom of wells. The corrosion might indicate intolerance of polystyrene to these solvent combinations, although the solvent was completely evaporated in a very short time (<10 min). The test of lipid measurement using chloroform only did not show any corrosion, however, a very thick and sticky layer formed when sulfuric acid was added, and color development was less than tests of lipid in other solvent combinations (Fig. 2). This phenomenon was also observed when the assay was performed in polypropylene microtubes, which suggests that the lipid layer might form because of over-drying when the solvent is evaporated rather than solvent intolerance of the assay container. Although the thick and sticky lipid layer could be dissolved after intensive mixing, this would not be suitable for an assay in a microplate where intensive mixing may be difficult to perform.

Concentration of Vanillin and Phosphoric Acid

Four different vanillin concentrations were tested based on the results from the preliminary test described earlier. When concentrated sulfuric acid addition was fixed at 100 μl , all concentrations of vanillin in 68% phosphoric acid tested in this study gave an even linearity of absorbance at 540 nm in the range 7.5–120 μg lipid (Fig. 3). In order to balance the conservation of vanillin use and sufficiency of absorbance, 0.25 mg/ml vanillin in 68% phosphoric acid was selected for the following optimization of phosphoric acid concentration. The purpose of testing different phosphoric acid concentrations was to investigate the possibility of using a lower amount of phosphoric acid while retaining linearity and sufficient absorbance. According to a previous report [18], when phosphoric acid concentration was varied between 17.6 and 70.4%, the

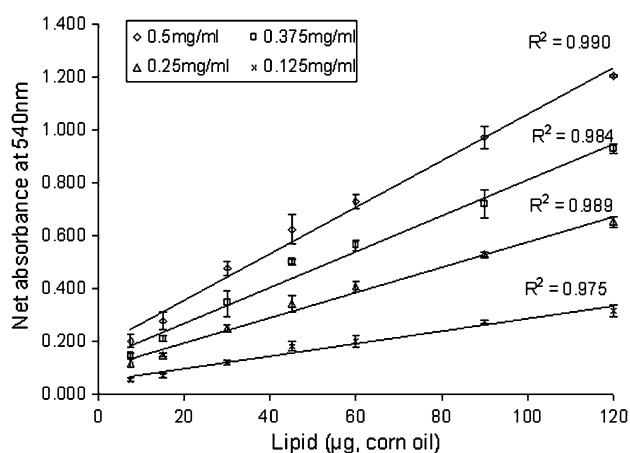


Fig. 3 Influence of vanillin concentration in vanillin–phosphoric acid reagent on color development. The vanillin–phosphoric acid reagent was prepared with 68% phosphoric acid. Samples were incubated with 100 μ l concentrated H_2SO_4 at 90 $^\circ\text{C}$ for 20 min. 100 μ l vanillin–phosphoric acid reagent was added for color development. Absorbance measurements were made after 5 min of color development. Data points and error bars represent the mean and standard deviation of three replicate samples

higher concentration provided a greater and more stable color response. Thus, four different concentrations in this range were selected for testing. Figure 4 shows that lower phosphoric acid concentration gave a higher absorbance. This is in contrast to previous reports. The result suggests that the water content in the reagent might play a role in promoting the color response and an increase in the stability. Based on this result, the concentration of vanillin was further optimized. Figure 5 shows that the reduced

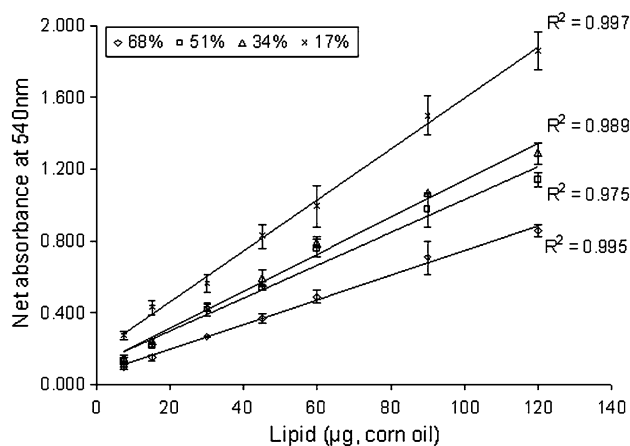


Fig. 4 Influence of phosphoric acid concentration in vanillin–phosphoric acid reagent on color development. The vanillin concentration was fixed at 0.25 mg/ml. Samples were incubated with 100 μ l concentrated H_2SO_4 at 90 $^\circ\text{C}$ for 20 min. 100 μ l vanillin–phosphoric acid reagent was added for color development. Absorbance measurements were made after 5 min of color development. Data points and error bars represent the mean and standard deviation of three replicate samples

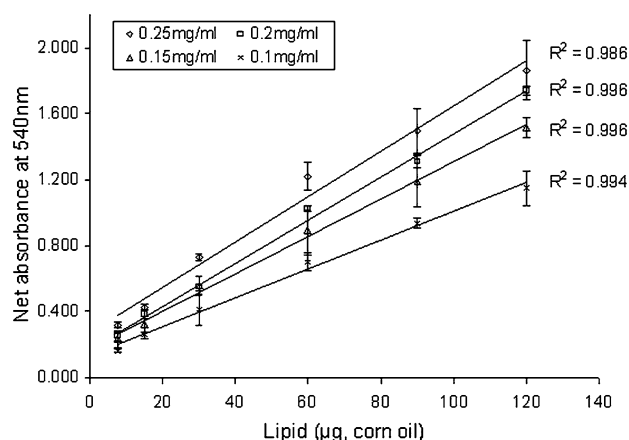


Fig. 5 Optimization of vanillin concentration in vanillin–phosphoric acid reagent for color development. The vanillin–phosphoric acid reagent was prepared with 17% phosphoric acid. Samples were incubated with 100 μ l concentrated H_2SO_4 at 90 $^\circ\text{C}$ for 20 min. 100 μ l vanillin–phosphoric acid reagent was added for color development. Absorbance measurements were made after 5 min of color development. Data points and error bars represent the mean and standard deviation of three replicate samples

vanillin concentration in 17% phosphoric acid provided equivalent absorbance linearity at 540 nm. Thus, vanillin concentration at 0.1 mg/ml was selected for the next study.

Volumes of Vanillin–Phosphoric Acid Reagent and Concentrated Sulfuric Acid

In an attempt to reduce water use associated with the vanillin reagent, four volumes containing an equivalent amount of vanillin in 17% phosphoric acid were examined. Volumes of 100 μ l (0.1 mg/ml) and 50 μ l (0.2 mg/ml) provided fairly even linearity of absorbance compared to 20 μ l (0.4 mg/ml) and 25 μ l (0.5 mg/ml) (Fig. 6).

The volume of concentrated sulfuric acid might affect the reaction quality. In one report, the results were more consistent when the volume of concentrated sulfuric acid was more than 100 μ l [25]. Therefore, sulfuric acid volumes of 100, 150 and 200 μ l were examined to determine the effect of sulfuric acid loading volume on the color response and stability. Figure 7 shows that higher sulfuric acid volume did not result in better linearity and color development. The results suggest that a moderate ratio between water and sulfuric acid is important for color development.

Reaction Time and Stability of Absorbance Linearity

In the published format of the micro-scale assay, samples were incubated with concentrated sulfuric acid at 100 $^\circ\text{C}$ for 10 min. Previous reports also indicated that the heating temperature will affect the color response [19]. Because of

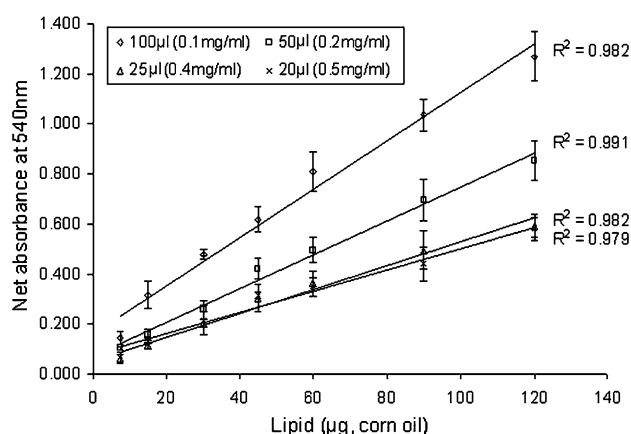


Fig. 6 Influence of the volume of vanillin-phosphoric acid reagent for color development. Vanillin-phosphoric acid reagents were prepared in 17% phosphoric acid. Samples were incubated with 100 μ l concentrated H_2SO_4 at 90 $^{\circ}C$ for 20 min. Vanillin-phosphoric acid reagent was added for color development. Absorbance measurements were made after 5 min of color development. Data points and error bars represent the mean and standard deviation of three replicate samples

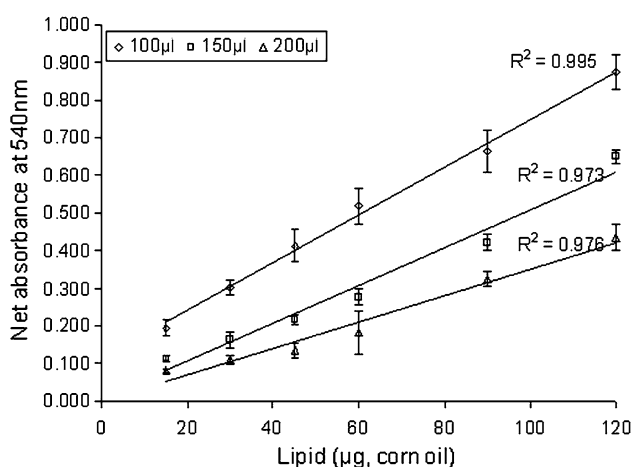


Fig. 7 Influence of volume of concentrated sulfuric acid for color development. Samples were incubated at 90 $^{\circ}C$ for 20 min. 50 μ l Vanillin-phosphoric acid reagent (0.2 mg vanillin per ml 17% phosphoric acid) was added for color development. Absorbance measurements were made after 5 min of color development. Data points and error bars represent the mean and standard deviation of four replicate samples

the temperature limitation of polystyrene, the reaction temperature was set at 90 $^{\circ}C$ and longer incubation times were used to compensate for the temperature difference. Figure 8a shows that at a reaction time of 20 min, the absorbance increased with color development time. Linearity in absorbance was stable for 3.5 h and the results from different reaction times showed a similar pattern. While longer reaction times produced higher background absorbance (Data not shown), the total absorbance was only slightly increased (Fig. 8b). Since color develops

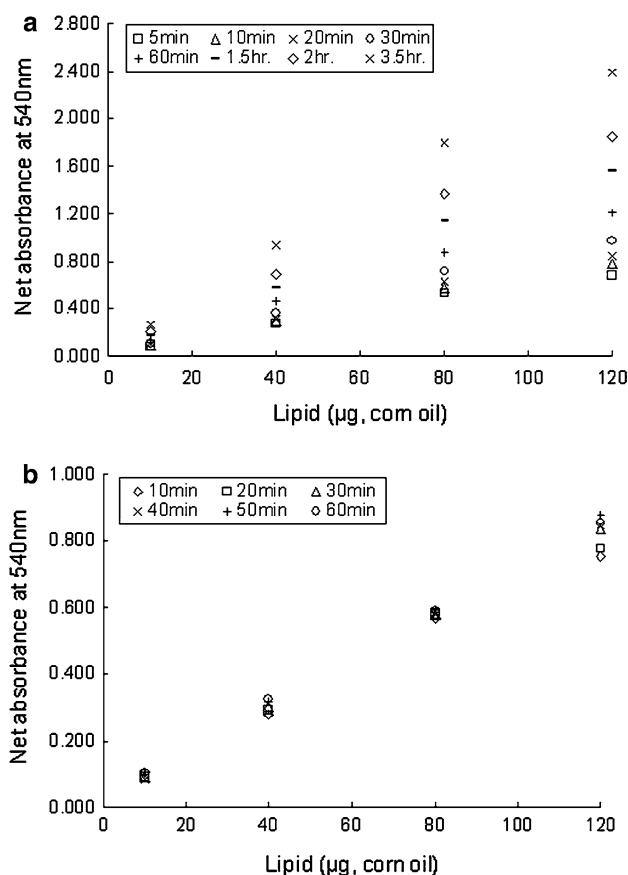


Fig. 8 Test of reaction time and stability for color development. **a** 20 min reaction time with color development from 5 min to 3.5 h, **b** 10–60 min reaction time with 10 min color development. 50 μ l vanillin-phosphoric acid reagent (0.2 mg vanillin per ml 17% phosphoric acid) was added for color development. Data points represent the mean of three replicate samples

continuously, laborious transfer of the reaction mixture for absorbance readings causes considerable error when large numbers of samples need to be measured. In the developed assay, because the samples and standards are on the same sample plate and read at the same time, the timing error is minimized.

Relative Absorbance and Linearity of Pure Lipid Samples

Previous reports indicated that selecting an appropriate standard was important for assessing lipid content in different types of samples [18, 19, 29]. Thus, different types of oil and cholesterol were tested using the new assay format (Fig. 9). The slope of the standard curve was significantly affected by lipid type ($p < 0001$). Standard slopes for cod liver oil and cholesterol were significantly lower than slopes for plant oils (Table 1). There were no significant differences in slopes among the plant oils, except flax oil which was different from all other oils

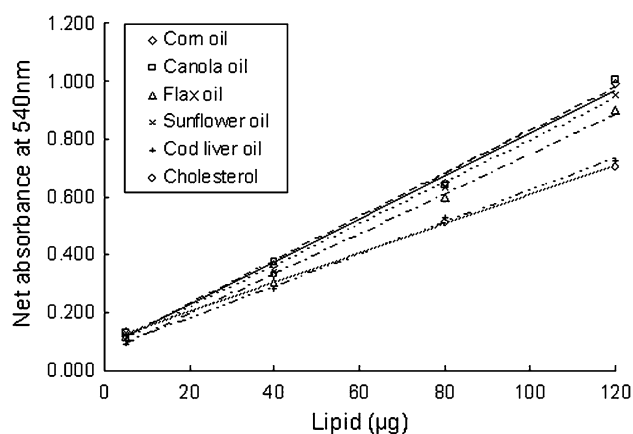


Fig. 9 Absorbance linearity of different lipid types. Samples were incubated with 100 μ l concentrated H_2SO_4 at 90 $^{\circ}C$ for 20 min. 50 μ l vanillin–phosphoric acid reagent (0.2 mg vanillin per ml 17% phosphoric acid) was added for color development. Absorbance measurements were made after 10 min of color development. Data points represent the mean of three replicate samples

Table 1 Mean slopes from standard curves measured using developed method

Lipid type	Mean slope (Au/ μ g)*
Canola oil	0.00755 A
Corn oil	0.00746 A
Sunflower oil	0.00721 AB
Flax oil	0.00685 B
Cholesterol	0.00502 C
Cod liver oil	0.00554 C

* Means followed by the same letter within columns are not significantly different according to Tukey's HSD means comparison test ($n = 3$), $p < 0.05$

except sunflower oil. Cod liver oil and cholesterol were also statistically equivalent (Table 1). Like other methods, standard selection should be based on the source of the sample (i.e., plant or animal) for the microplate method.

Final Microplate Procedure

The final procedure involves the addition of <100 μ l of samples and standards that contain 5–120 μ g lipids directly into the bottom of microplate wells. After the solvent is evaporated at 90 $^{\circ}C$, 100 μ l of concentrated sulfuric acid is added to each well. Then, the microplate is incubated at 90 $^{\circ}C$ for 20 min. Background absorbance at 540 nm is measured when the microplate is cooled to room temperature on ice water (~ 2 min). 50 μ l of vanillin–phosphoric acid reagent (0.2 mg vanillin per ml 17% phosphoric acid) is added to each well for color development for 10 min, and then absorbance is measured at 540 nm. Since the heating and cooling steps will affect the absorbance, evenly

heating and cooling the reaction mixtures are critical for having good absorbance linearity. Thus, performing the incubation in a microplate dry heating module or on a water bath with careful handling is recommended.

Applicability of the Assay for Measuring Solvent Extracted Lipid from Microalgae

Since the SPV method has been reported to have a variable response to varying sources of lipids [30], lipids extracted from four different strains of green microalgae were analyzed using the macro-gravimetric method and compared to the new microplate assay. Corn oil was used as the standard for the colorimetric method because microalgal lipids usually have a higher unsaturated ratio and are relatively close in composition to vegetable oil [16, 31–34]. The determined total lipids were converted to lipid percentage of dry algal biomass. Figure 10a shows the correlation of lipid percentage between the microplate format of the SPV method and the macro-gravimetric

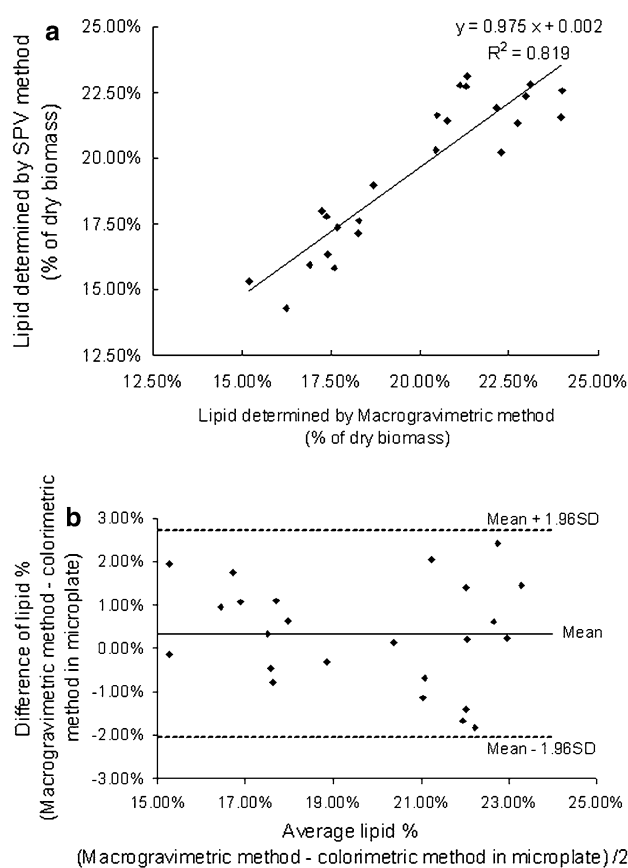


Fig. 10 **a** The correlation between lipid percentage of algae determined using the microplate colorimetric method and macro-gravimetric method. **b** Difference against average of macro-gravimetric and microplate format colorimetric measurement of lipid percentage with 95% limits of agreement (dashed lines)

method; there is a strong correlation between the macro-gravimetric method and the microplate format SPV method ($R^2 = 0.819$, $F = 99.72$, $p < 0.0001$). Figure 10b shows that the mean difference and standard deviation are 0.33 and 1.21% (test pairs = 24), respectively, and all differences fall between the 95% limits of agreement. Moreover, the matched paired t test is 1.35 ($p = 0.19$, two tails), which provides evidence that there is no significant difference between the microplate format of the SPV method and the macro-gravimetric method. This result is consistent with an earlier report [29] and suggests that the new SPV method in a microplate format can be applied for the quantification of total lipids from different types of microalgae samples with proper standard selection.

Conclusion

Total lipid quantification is frequently performed in many research areas with different types of samples. The present work reports on a modified colorimetric method for quantitative analysis of total lipid using a high-throughput microplate format. The presented method uses extracted and purified lipid samples. The extraction step minimizes interferences associated with other components in the sample and allows different samples from a variety of research areas to be analyzed using the same conditions. This approach could be very useful when screening the lipid content in different cell cultures and plants. With the development of high-throughput cell disruption technologies such as 96-well plate bead beaters and microwave extraction [35], lipid extraction from large numbers of samples could be done in minutes.

The new assay method possesses many advantages: (1) it requires a small amount sample and the sample volume can be adjusted to fit in the standard range, (2) it requires less time (<1 h) and less labor when a large number of samples is analyzed, and (3) the color development is more consistent with modifications of reagent concentrations. Moreover, the reagents are confined in the microplate through the entire assay procedure, which enables fast and accurate reading of multiple samples. In the final procedure, it is recommended that the volume of the sample should be less than 100 μ l in order to ensure a complete reaction with the sulfuric acid. In addition, uniform heating and cooling are important for consistency in the reaction. The method was validated against the macro-gravimetric method. When corn oil was used as a standard, statistically equivalent results were achieved when both methods were used to measure total lipids in four different strains of *Chlorella*.

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